

Configuration-as-Substrate-Content Across All Paper 3 Claims

Derivation Note D1.24 — Note #489 in the CKS Defensive-Publication Series

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Paper 3 Claim 5 establishes configuration as substrate content with recursive applicability. This note formalizes the third of four Claim 5 sub-commitments (D1.24): the architectural commitment that Claim 5’s configuration-as-substrate-content principle applies not only to the six FAI configuration dimensions (D1.22) and to configuration-of-configuration (D1.23), but to every architectural object named in every other Paper 3 claim. Claim 5 is simultaneously a standalone architectural commitment and a unifying statement that retroactively governs Claims 1 through 4 and Claim 6. The note states the sub-commitment precisely, applies it to each claim area with concrete configuration objects, states its prior-art significance in closing adversarial gaps at every claim boundary, and traces its inheritance from Paper 2 Claim 6’s parallel recursive-applicability commitment. Three failure modes are identified and an operational test is provided for each of the five other claim areas.

1. The sub-commitment stated

D1.24 formalizes the following architectural commitment:

“The configuration-as-substrate-content principle asserted by Paper 3 Claim 5 applies to every architectural object named in every other Paper 3 claim. For each of Claims 1, 2, 3, 4, and 6, the configuration specifications governing how that claim’s architectural mechanism operates — how it is constructed, parameterized, and governed — are authored substrate content subject to the same governance authority that governs configuration substrate at FAI-event scope. No claim area has a special configuration mechanism outside the substrate-content register.”

This sub-commitment has two distinct logical components, both of which matter for prior-art purposes.

The first component is **universality**: Claim 5 is not scoped to FAI-event dimensions alone. Every Paper 3 claim presupposes that the governance of its mechanism is configured through authored substrate content. Claims 1 through 4 and Claim 6 each introduce an architectural mechanism (shared substrate construction, FAI operation, conflict handling, evolution-feed routing, population-scale propagation). For every such mechanism, the rules governing how the mechanism is configured are substrate content — not infrastructure defaults, not vendor-specific protocol invariants, and not institutional procedures external to the governance architecture.

The second component is **retroactive unification**: Claim 5 is the explicit statement of a principle that Claims 1 through 4 presuppose implicitly. By naming the principle explicitly, Claim 5 performs two functions simultaneously — it stands as a standalone prior-art claim for the recursive-applicability commitment itself, and it forecloses any adversarial argument that configuration governance in any specific claim area is architecturally novel.

2. Claim 5 as both standalone and unifying

The dual character of Claim 5 is what makes it architecturally important and what determines how this note relates to D1.22 (the six FAI configuration dimensions) and D1.23 (configuration-of-configuration recursion).

The standalone dimension is formalized in D1.22 and D1.23. D1.22 establishes that the six configurable FAI dimensions — sharing scope, cardinality, persistence policy, cooperation/competition variant, provenance carry-over depth, and multi-mediator coordination — are each authored substrate content subject to joint governance authority across participating Selves. Each dimension is independently claimable as a prior-art commitment. D1.23 establishes that the governance substrate specifying FAI configuration is itself substrate content: configuration-of-configuration is substrate content, with the recursion bottoming out at human-authored governance authority per Paper 1’s authority-versus-labor distinction.

The unifying dimension is formalized here in D1.24. The same principle that D1.22 applies to FAI dimensions and D1.23 applies to configuration-of-configuration applies to every architectural object across all six claims. Claims 1 through 4 and Claim 6 each carry a set of configuration objects — the rules specifying how their mechanisms operate. Claim 5 commits that all such configuration objects are substrate content. This is the architectural assertion that makes Claim 5 a unifying statement rather than merely a scoped sub-claim.

The consequence is that Claim 5 makes six independently claimable prior-art claims (one per FAI dimension) *and* makes a prior-art claim on the principle of cross-claim universality itself. An adversary who attempts to capture novelty at any specific claim boundary — “the governance of how the shared substrate is constructed is novel,” or “the governance of escalation routing is novel”

— encounters Claim 5 operating at the unifying register.

3. Application to each claim area

Claim 1 (Shared Substrate Construction)

Claim 1 establishes the shared substrate as the architectural object at the inter-Self perimeter — a perimeter-spanning substrate constructed temporarily for FAI events, carrying all six Paper 1 commitments within its scope, and dissolving at event conclusion. The configuration objects within Claim 1's scope include:

- **Construction-authorization specification:** which governance authority, operating under which conditions, is authorized to initiate shared substrate construction. This specification is authored substrate content within each participating Self's governance perimeter.
- **Initial orchestration rules:** the substrate-level rules governing what the shared substrate may carry at construction, how its content scope is initialized, and how participating Selves' home governance extends into the shared scope. These rules are authored substrate content per Paper 1's orchestration-rule authorship commitment, held at the joint authority of participating Selves.
- **Dissolution policy:** the specification of when and how the shared substrate dissolves and what happens to in-flight content at dissolution. This policy is authored substrate content, not a protocol invariant.

Claim 5's application to Claim 1 asserts that the governance of the shared substrate's existence as a governed object is itself governed through substrate content. The shared substrate is not merely a governed object that appears once constructed — the specification of how it comes to exist, what authority governs its construction, and how it terminates are all substrate content under Claim 5.

Claim 2 (FAI Operation)

Claim 2 establishes Full Aspect Integration as the canonical operation over the shared substrate. The configuration objects within Claim 2's scope include the six FAI dimensions formalized in D1.22, which are not restated here. Additionally, at meta-specification scope, Claim 5 applies to:

- **Permissible-operation specification:** the meta-specification of which FAI operations are authorized for a given participating Self in a given organizational context. This specification — governing what the substrate permits, not just how a permitted operation is configured — is authored substrate content.
- **Aspect-exchange authorization:** which aspects of each participating Self's governance layer may be brought into an FAI event. The authorization mapping is authored substrate content under joint governance, not a structural property of the participating Selves' home architectures.

D1.22 covers the operational dimensions at FAI-event level. D1.24's contribution at Claim 2 scope is the meta-specification level — the governance of which FAI operations are available at all, which

is a configuration object distinct from the configuration of an individual FAI event.

Claim 3 (Conflict Handling)

Claim 3 establishes the three-tier inter-Self conflict-handling mechanism: preserve (substrate-level state preservation), resolve (orchestration-rule application with inspectable rules), and escalate (routing to joint governance authority). The configuration objects within Claim 3's scope include:

- **Tier-selection rules:** the authored orchestration rules governing which tier handles which conflict class. These rules — preserve for substrate-state conflicts, resolve for conflicts within orchestration-rule scope, escalate for conflicts requiring joint authority action — are authored substrate content per Paper 1 Claim 4 and Paper 3 Claim 3.
- **Conflict-class definitions:** the specification of which conflict types fall within which tier. These definitions are authored governance content, not inferred from the content of participating Selves' home substrates.
- **Escalation routing:** the specification of which governance authorities receive escalated conflicts — which human authority, operating at which perimeter, is the recipient of an escalated inter-Self conflict. This routing specification is authored substrate content; it is not determined by organizational chart or institutional procedure external to the governance architecture.
- **Recursive-governance depth at conflict scope:** how many layers of conflict-handling governance are themselves substrate content (a Claim 5 $\times$ Claim 3 intersection that D1.23 addresses at the general recursion level and Claim 3 instantiates at conflict-handling scope).

Claim 4 (Evolution Feed)

Claim 4 establishes the four-locus evolution-feed mechanism through which FAI dissolution events route content to each participating Self's home Paper 2 evolution machinery. The configuration objects within Claim 4's scope include:

- **Feed-eligibility specifications:** the authored governance content specifying which content classes produced by an FAI event are eligible for each evolution locus at a given participating Self's home perimeter — what is eligible for the action-feedback locus, what is eligible for DNA evolution locus, and what is excluded from home ingestion.
- **Hand-off boundary configuration:** the specification of how FAI dissolution triggers the hand-off to each participating Self's home evolution machinery. The hand-off boundary — where the shared substrate's dissolution ends and each home substrate's ingestion begins — is configured as substrate content, not hardcoded at implementation layer.
- **Per-mechanism ingestion policy:** the specification, at each participating Self's home perimeter, of how content arriving from FAI dissolution is processed by each of the three Paper 2 evolution mechanisms. These ingestion policies are authored substrate content at home perimeter scope, inheriting home Paper 2 governance shapes.

- **FAI-outcome routing:** the specification of which FAI event outcomes flow to which home substrates — relevant when an FAI event involves more than two participating Selves and routing is non-trivial. This routing specification is authored governance content.

Claim 6 (Population-Scale Propagation)

Claim 6 establishes population-scale collective evolution as the architectural object produced by accumulated FAI events across many participating Selves. At population scope, the configuration objects under Claim 5 include:

- **Propagation-scope specifications:** the governance content specifying what propagates across the inter-Self network at population scope — what accumulated FAI-event content is eligible for network-wide influence versus what remains within individual participating Selves’ home perimeters.
- **Specialization-preservation specifications:** the governance content specifying which specializations developed by individual participating Selves are preserved against homogenization pressure at population scope. These specifications are substrate content within each participating Self’s governance perimeter.
- **Cross-organizational propagation authority:** the governance specification of which authority governs propagation decisions that cross organizational boundaries. At population scope, joint authority extends to propagation governance; this authority architecture is authored substrate content, not determined by external federation agreements.

4. Prior-art significance: closing the adversarial gap at every claim boundary

The prior-art significance of D1.24 is structural. For each of Claims 1, 2, 3, 4, and 6, an adversary seeking to establish novelty at the configuration-governance layer might argue: “The architectural mechanism described in Claim N exists in the prior art, but the specific governance architecture specifying how that mechanism is configured is novel.” This adversarial argument attempts to find unoccupied territory at the configuration-governance boundary of each individual claim.

Claim 5’s cross-claim application closes every such gap simultaneously. The argument fails for Claim 1 (construction-authorization and dissolution-policy as substrate content are claimed under Claim 5). It fails for Claim 2 (permissible-operation specification as substrate content is claimed under Claim 5). It fails for Claim 3 (tier-selection rules, conflict-class definitions, and escalation routing as substrate content are claimed under Claim 5). It fails for Claim 4 (feed-eligibility specifications, hand-off boundary configuration, and ingestion-policy specifications as substrate content are claimed under Claim 5). And it fails for Claim 6 (propagation-scope and propagation-authority specifications as substrate content are claimed under Claim 5).

The architectural reason this works is that Claim 5 is not a collection of claim-specific governance sub-claims. It is a single architectural commitment that applies uniformly: any configuration specification governing any architectural object in Paper 3’s architecture is substrate content subject

to the same governance authority. There is no special mechanism for conflict-handling configuration, no special mechanism for evolution-feed configuration, no special mechanism for shared-substrate-construction configuration. The uniformity of the principle is what closes the adversarial gap uniformly across all claim areas.

5. Inheritance from Paper 2 Claim 6

The cross-claim recursive applicability of D1.24 is the Paper 3 analog of Paper 2 Claim 6 (B0.06 / B1.20). The parallel is precise enough to be worth stating explicitly, because it reveals a trilogy-level architectural pattern.

Paper 2 Claim 6 established the enterprise brain Self as architecturally coherent when Paper 1's six commitments apply recursively at every scope level — at the cell level, at the aspect level, and at the Self level. Paper 2 Claim 6's contribution was to assert that Paper 1's commitments are not scoped to any particular structural level; they apply wherever substrate-mediated coordination operates. This made Claim 6 both a standalone claim about the enterprise brain Self pattern and a unifying statement that Paper 1's architecture extends coherently to all scope levels Paper 2 introduces.

Paper 3 Claim 5 performs the same logical move at Paper 3's scope. Claim 5 asserts that the configuration-as-substrate-content principle is not scoped to the FAI-event level or to any particular claim's mechanism; it applies to every architectural object at every claim boundary Paper 3 introduces. This makes Claim 5 both a standalone claim about configuration-as-substrate-content with recursive applicability (D1.22, D1.23) and a unifying statement that the configuration principle extends coherently to all claim areas Paper 3 introduces (D1.24).

The pattern is consistent across the trilogy: each paper's extension claim establishes that the prior architecture's governing principles apply recursively across the new scope the current paper introduces. This pattern is itself a prior-art commitment: any downstream architecture that adopts the recursive-applicability structure for configuration governance inherits from the trilogy's explicit articulation of that pattern.

6. Three failure modes

D1.24 defends against three specific failure modes at the cross-claim application level.

Failure mode 1 — Claim-specific configuration escape. This failure mode occurs when a deployment treats Claim 5 as applying to FAI-event dimensions (Claim 2 scope) but not to the configuration of other claim areas. In this failure, the orchestration rules governing conflict-tier selection, or the specification of which governance authority receives escalated conflicts, or the hand-off boundary configuration for evolution feed, are implemented as infrastructure defaults or vendor-specific protocol settings rather than as authored substrate content. The system is partially Claim 5 compliant — it authors FAI configuration as substrate content — but leaves claim-specific

configuration outside the substrate-content register. The failure is detectable: configuration objects described in §3 of this note are not findable as authored substrate content with governance authorization.

Failure mode 2 — Selective Claim 5 scope. This failure mode occurs when an implementation argues that Claim 5 governs “configuration” in a narrow sense (e.g., the six FAI dimensions) but that the governance of conflict handling or evolution-feed routing is a “mechanism,” not “configuration,” and therefore outside Claim 5’s scope. The architectural response is that Claim 5’s scope is defined by what constitutes a configuration object for a given architectural mechanism, not by the word “configuration” applied narrowly. The tier-selection rules that govern resolve vs. escalate are configuration objects for the conflict-handling mechanism; the feed-eligibility specifications are configuration objects for the evolution-feed mechanism. The scope of Claim 5 is the scope of authored governance specifications for any Paper 3 mechanism.

Failure mode 3 — Novel governance mechanism per claim area. This failure mode occurs when a designer introduces a special governance mechanism for a specific claim area — a dedicated conflict-tier governance module, a separate evolution-feed authorization framework, a distinct shared-substrate-construction approval workflow — each implemented outside the substrate-content register. The claim of architectural novelty for each such mechanism is exactly what Claim 5’s cross-claim application forecloses. The architectural commitment is that each claim area’s configuration is governed as substrate content through the same governance authority that governs the rest of the substrate — no special mechanisms, no separate authority architectures per claim.

7. Operational test

For each of the five other claim areas, the following test applies. The test asks: what should be findable as authored substrate content if Claim 5’s cross-claim application is correctly instantiated?

Claim 1 (Shared Substrate Construction): An observer should be able to locate, within each participating Self’s governance substrate, an authored specification — with identifiable governance authorization — of: (a) the conditions under which this Self may participate in shared substrate construction, (b) the initial orchestration rules the Self contributes to the shared substrate at construction, and (c) the dissolution policy governing end-of-event behavior. These should be distinct from any infrastructure default or vendor-supplied protocol setting.

Claim 2 (FAI Operation): An observer should be able to locate an authored specification of: (a) which FAI operation types are permissible for this Self in the current organizational context, and (b) which aspects of this Self’s governance layer are authorized for exchange in an FAI event. These meta-specifications should exist as authored substrate content, separate from the event-level configuration objects covered by D1.22.

Claim 3 (Conflict Handling): An observer should be able to locate authored orchestration rules specifying: (a) which conflict classes are routed to preserve, resolve, and escalate tiers respectively, (b) the definitions of conflict classes used in those routing rules, and (c) the identity of governance authorities who receive escalated conflicts. All three should be authored substrate content with governance authorization, not derived from institutional hierarchy or organizational structure external to the substrate.

Claim 4 (Evolution Feed): An observer should be able to locate an authored specification of: (a) content-class eligibility for each evolution locus at this Self’s home perimeter, (b) the hand-off boundary configuration specifying when dissolution triggers ingestion, and (c) per-mechanism ingestion policies. These should be authored substrate content at this Self’s home perimeter, distinct from the shared substrate content that governs the FAI event itself.

Claim 6 (Population Propagation): An observer should be able to locate authored specifications of: (a) what this Self permits to propagate across the inter-Self network at population scope, (b) which of this Self’s specializations are marked for preservation against population-scale homogenization, and (c) what authority governs cross-organizational propagation decisions in which this Self participates. These should be substrate content within this Self’s governance perimeter, not determined by external network protocol.

A deployment that passes all five of these tests, in addition to the FAI-dimension tests from D1.22 and the configuration-of-configuration test from D1.23, instantiates Claim 5’s full sub-commitment structure through D1.24.

8. Conclusion

D1.24 formalizes the cross-claim universality of Claim 5’s configuration-as-substrate-content principle. Claim 5 is not scoped to FAI-event dimensions; it is the explicit assertion of a principle that Claims 1 through 4 and Claim 6 presuppose. For each of the five other claim areas — shared substrate construction, FAI operation, conflict handling, evolution feed, and population-scale propagation — the configuration objects governing how the claim’s mechanism operates are authored substrate content subject to the same governance authority.

This sub-commitment performs three architectural functions simultaneously. It gives Claim 5 its unifying character alongside its standalone character (§2). It closes the adversarial gap that would otherwise exist at each individual claim boundary (§4). And it instantiates the trilogy-level pattern — established at Paper 2 Claim 6 scope and extended here — of each paper’s extension claim asserting recursive applicability of the prior architecture’s governance principles across all scope levels the current paper introduces (§5).

The operational consequence is that a correctly instantiated Paper 3 architecture has no claim-specific configuration mechanisms: one substrate-content governance architecture covers configuration at every level, at every claim boundary, for every mechanism Paper 3 introduces.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

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